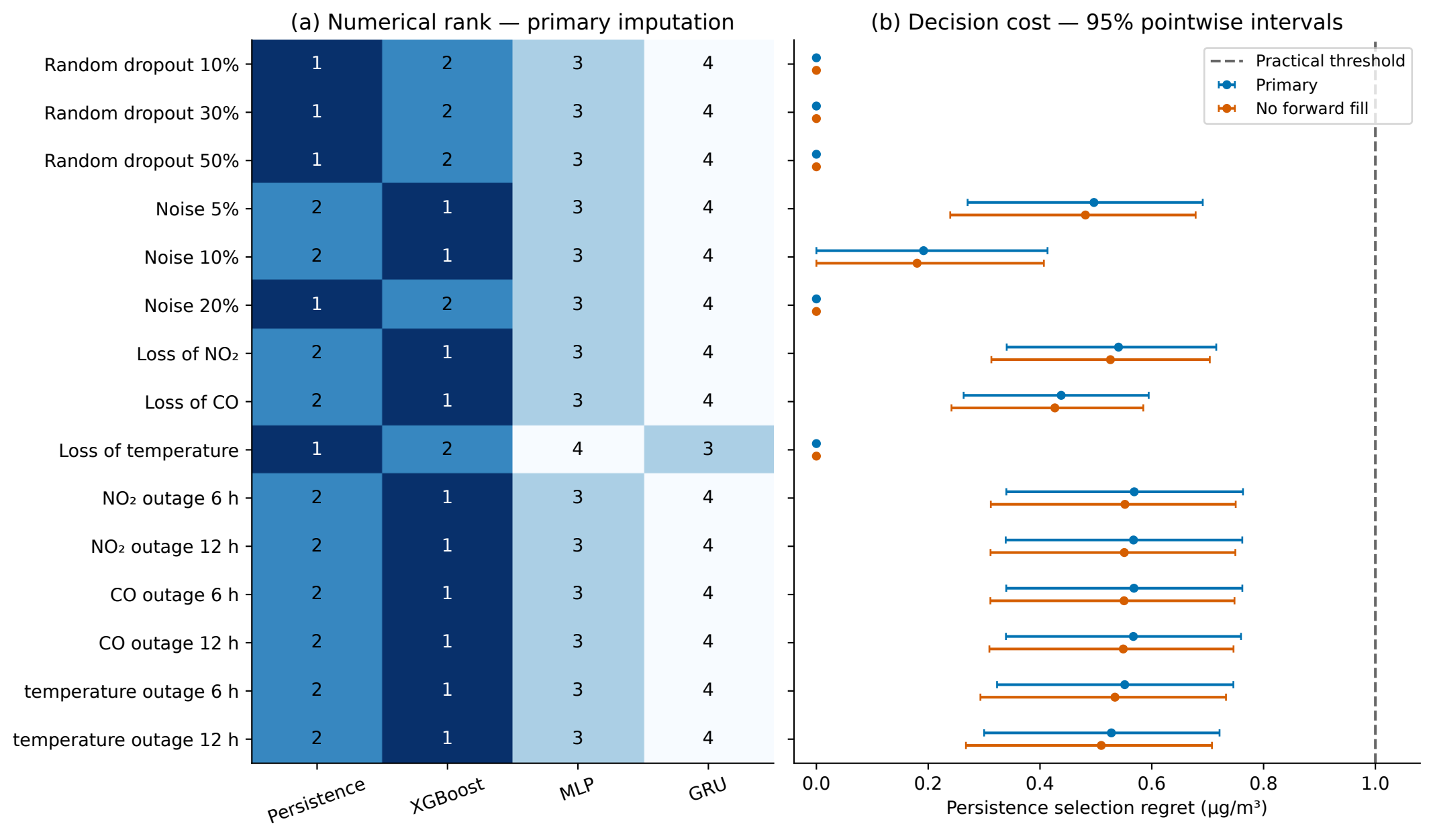
(a) Numerical rank — primary imputation
Random dropout 10%
Random dropout 30%
Random dropout 50%
Noise 5%
Noise 10%
Noise 20%
Loss of NO₂
Loss of CO
Loss of temperature
NO₂ outage 6 h
NO₂ outage 12 h
CO outage 6 h
CO outage 12 h
temperature outage 6 h
temperature outage 12 h
Persistence
XGBoost
MLP
GRU
(b) Decision cost — 95% pointwise intervals
Practical threshold
Primary
No forward fill
0.0
0.2
0.4
0.6
0.8
1.0
Persistence selection regret (µg/m³)